

FPGA Tick-to-Trade — Implementation Results

June 2026

Synthesis & place-and-route results, with a baseline for tracking optimisation.

This document records real Vivado synthesis + implementation (placed & routed) results for `tick_to_trade_top`, so successive optimisations can be compared. Reports live in `results/synthesis/` and `results/implementation/`.

Run B0 — Baseline (registered order-book array)

Item	Value
Target device	xcu200-fsgd2104-2-e (Alveo U200 / UltraScale+ VU9P — AWS F1 class)
Tool	Vivado 2025.x, out-of-context (no physical pin constraints)
Target clock	200 MHz (5.000 ns period)
Flow	synth_design → opt → place → route (fully routed)

Timing — **FAIL** at 200 MHz

Metric	Value
Worst Negative Slack (WNS)	-2.373 ns
Failing endpoints	41,206
Total Negative Slack (TNS)	-51,387 ns
Max achievable clock	≈ 135 MHz (1/(5.0+2.373) ns)
Implied tick-to-trade	39 cycles × 7.37 ns ≈ 290 ns (not 195 ns)

Utilisation (placed) — tiny footprint

Resource	Used	Available	%
CLB LUTs	36,057	1,182,240	3.05
CLB Registers	37,672	2,364,480	1.59
CARRY8	265	147,780	0.18
Block RAM Tile	0	2,160	0.00
DSPs	0	6,840	0.00

Power & DRC

Total on-chip power	2.844 W (dynamic 0.371 W, static 2.473 W; low confidence)
Routing	fully routed (52,563 / 52,563 routable nets)
DRC	2 critical warnings: NSTD-1, UCIO-1 (unconstrained I/O — expected in OOC flow with no pin constraints; benign for F1-shell integration)

Critical path & root cause

Source	u_book/entries_reg[7] [shares] [8]
Destination	u_book/entries_reg[219] [shares] [18]
Data delay	7.036 ns — route 5.969 ns (85%) , logic 1.067 ns, 11 logic levels

Root cause: the order-book `entries[]` array (256×130 bits) synthesised to **37 K flip-flops (BRAM = 0)** instead of memory, and the RESCAN FSM reads *all* entries combinatorially. The huge fan-out of that register file makes the path **route-dominated (85%)**, so it misses 200 MHz badly. Logic is fine (<1.1 ns); the problem is physical (FF array + combinational scan).

Planned fix — Run B1 (BRAM-based order book)

Convert `entries[]` to a true **BRAM with synchronous (registered) reads**. The RESCAN FSM already walks one entry per cycle, so a 1-cycle BRAM read drops in naturally; the Add fast-path stays $O(1)$ (writes + level-insert from the message, no read). This removes both the 37 K-FF array and the route blow-up and should close 200 MHz.

Correction 1 — Run B1 (BRAM order book, measured)

The BRAM order book was implemented and re-run through synth + implementation on the VU9P. **Outcome: the resource/route problem is fixed, but timing still fails — for a new reason.**

Metric	B0 (FF array)	B1 (BRAM)	Change
CLB Registers (FF)	37,672	4,308	−89% ✓
CLB LUTs	36,057	6,790	−81% ✓
Block RAM Tile	0	2	inferred ✓
DSPs	0	0	—
Failing endpoints	41,206	3,612	−91% ✓
Total Negative Slack	−51,387 ns	−2,522 ns	−95% ✓
Worst Negative Slack	−2.373 ns	−4.563 ns	worse
Total on-chip power	2.844 W	2.607 W	−8%
Timing @ 200 MHz	FAIL	FAIL	not yet closed

What worked: `entries[]` now infers block RAM (2 tiles) instead of 37 K flip-flops; LUTs/FFs collapsed ~85%; the route-dominated FF-array path is gone — failing endpoints and total violation both dropped ~90–95%. The original root cause is resolved.

New critical path (B1):

Source	<code>u_book/bid_lv_reg[cnt][0]</code> (level cache)
Destination	<code>risk_reason[1]</code> (output pin / OBUF)
Data delay	5.168 ns — logic 2.494 ns (48%), route 2.673 ns (52%), 14 levels, 5×CARRY8

Why WNS got worse: this is no longer a routing problem — it is a long *combinational* chain: book level cache → strategy (depth-weighted cross-multiply) → `risk_check` (price-collar subtract, position add) → straight to the decision/`risk_reason` output pins, with no register in between. `risk_check` was added as a *zero-latency combinational* gate (to preserve 195 ns); once the FF-array path was removed, this chain became the worst path.

Correction — Run B2 (register the risk/decision stage), measured

Made `risk_check` a **1-cycle pipeline** (registered outputs) and registered the decision output ports, splitting book → strategy → risk → pin. Cost: **+1 cycle** (39 → 40 cycles, 195 → **200 ns**). `tick_to_trade_top` was re-synthesised + implemented on the VU9P:

Metric	B1 (comb. risk)	B2 (registered)	Change
WNS	−4.563 ns	−2.217 ns	+2.35 ns ✓
Failing endpoints	3,612	87	−3,525 (98% fewer)
Worst-path dest.	<code>risk_reason</code>	<code>s_axil_awready</code>	moved off the trade path
Worst-path logic	14 levels	2 levels (LUT2+OBUF)	—

B2 closed the trading datapath. The original decision cone (book $\rightarrow$ strategy $\rightarrow$ risk $\rightarrow$ pin) no longer appears anywhere in the failing list. The residual -2.217 ns is no longer a logic problem — it is a **clock-insertion + I/O-budget artifact of the out-of-context run**:

- The worst path (`s_axil_bvalid_reg` $\rightarrow$ `s_axil_awready`, the AXI-Lite latency-readout port) has only **2 logic levels** but rides a **2.714 ns clock-insertion delay** — one global BUFG fanning out to 4,757 loads, i.e. 54% of the 5 ns period gone before any logic.
- Insertion delay cancels on internal register $\rightarrow$ register paths (which meet 200 MHz) but is charged in full against I/O paths. With the former 40% (2.0 ns) I/O budget the boundary was unsatisfiable even though the logic is fine.

Fix (constraints, not RTL): in `rtl/top.xdc` — (1) `set_false_path` the whole `s_axil_*` boundary (a slow, host-pollled status port, not a 200 MHz datapath; the Shell closes its timing in-context); (2) right-size the boundary I/O budget 40% $\rightarrow$ 10% (0.5 ns) to reflect that the real F1 build places the clock in clock regions (insertion ~ 0.3 ns) and the Shell registers the CL/SH interface. Re-run synth+impl to confirm; the internal logic already meets 200 MHz.

Correction 2 — Multi-symbol engine ($\times 4$), measured

`multi_symbol_top` (NSYMBOLS=4: four per-symbol books routed by `stock_locate`) was synthesised + implemented on the VU9P.

Metric	single-sym B1	multi-sym ($\times 4$)	Note
Block RAM Tile	2	8	exactly $4\times$ — one BRAM book/symbol $\checkmark$
CLB LUTs	6,790	19,942	$\sim 3\times$ (shared strategy/risk)
CLB Registers (FF)	4,308	8,984	$\sim 2\times$
WNS	-4.563 ns	-5.362 ns	worse
Failing endpoints	3,612	10,899	—
Power	2.607 W	2.869 W	—
Routing	clean	fully routed, 0 errors	$\checkmark$

Confirms the BRAM book scales: 4 symbols $\rightarrow$ 8 BRAM tiles (4×2), no FF explosion. **Worst path** (`gbook[3] bid_lv` $\rightarrow$ `m_axis_tvalid`, 18 levels, $8\times$ CARRY8) is the same combinational book $\rightarrow$ strategy $\rightarrow$ risk $\rightarrow$ pin chain, *lengthened* by the 4-way active-symbol output mux. This run pre-dates the register fix; the fix has since been applied to `multi_symbol_top` too (re-run pending).

Correction 3 — Run B3 (pipeline strategy + collar reference)

Analysis of the routed B1/M1 reports showed that registering only the output (B2) is necessary but *may not be sufficient*: two structural problems remain on the decision cone.

Residual issue (from B1/M1 routed paths)	Why it still hurts after B2
Book $\rightarrow$ <code>mid_price</code> (add+shift) $\rightarrow$ <code>risk_check</code> collar $\rightarrow$ output.	The collar reference was a live combinational function of the book best price, feeding the risk comparators.
Strategy $\Sigma(\text{weight}\cdot\text{size})$ and the 64-bit cross-multiply in one cycle.	Weighted-sum + cross-multiply + compare were one deep cone (route 52–58%).

Fix (B3):

- **strategy_imbalance** is now a **2-stage pipeline**: stage 1 registers the depth-weighted volumes w_{bid}/w_{ask} and the book snapshot; stage 2 does the cross-multiply + lot sizing + fire decision. This isolates the wide summation from the cross-multiply compare.
- Both tops register the collar reference (book mid) **2 deep** (`mid_r1/mid_r2`) so the risk comparators start at a flip-flop, not the `book` → `mid` adder, and stay aligned with the now 2-cycle strategy. (Bonus: this also removes a 1-cycle order/mid skew that existed in B2.)

Cost: +1 cycle (tick-to-trade 40 → 41 cycles, 200 → **205 ns**). The split halves the longest combinational segment on both the strategy and risk sub-paths.

Status: implemented and verified in simulation — **95/95 cocotb tests pass**, lint clean, tick-to-trade measured at **205 ns**, multi-symbol decisions still correctly attributed. *Not yet re-synthesised* — re-run synth+impl on `tick_to_trade_top` (then `multi_symbol_top`) to populate the real B3 timing and confirm 200 MHz closure.

Correction 4 — Run B4 (XDC: false-path AXI-Lite + 10% I/O budget), measured

Re-synthesised + implemented `tick_to_trade_top` with the B4 constraints fix. The AXI-Lite false-path + 10% I/O budget corrected most of the B2 I/O violations:

Metric	B2	B4	Change
WNS	−2.217 ns	−0.578 ns	+1.64 ns ✓
Failing endpoints	87	47	−40
Worst-path logic	2 levels (AXI-Lite)	1–2 levels (FF→OBUF)	all FF-to-output pins
Internal reg→reg	meets 200 MHz	meets 200 MHz	confirmed

Remaining **47** violations are all **FF→OBUF** paths with **0–1** logic levels:

Destination (output pin)	Source (FF)	Levels	Slack
<code>s_axis_tready</code>	<code>u_book/state_reg[0]</code>	2 (LUT3+OBUF)	−0.578 ns
<code>risk_reason[0]</code>	<code>risk_reason_reg[0]</code>	1 (OBUF)	−0.461 ns
<code>m_axis_tdata[*]</code>	<code>m_axis_tdata_reg[*]</code>	1 (OBUF)	~−0.44 ns

Root cause (unchanged): OOC BUFG clock insertion 2.658 ns across fanout ~4,752 loads. This delay is charged in full on FF→output-pin paths even after the I/O budget reduction, because the OBUF propagation (~1.1 ns) + insertion (2.658 ns) ≈ 3.76 ns exceeds the 0.5 ns required budget. In the real F1 build the Shell registers every CL/SH boundary port and the clock is placed in clock regions (insertion ~0.3 ns); these paths do not exist in-context.

Fix (B5): false-path *all* remaining CL/SH boundary ports (`s_axis_*`, `m_axis_*`, `risk_*`, `halt`) so STA evaluates only internal register→register paths, which already meet 200 MHz. The `top.xdc` has been updated accordingly.

Correction 5 — Run M2 (multi-symbol B4 constraints fix), measured

`multi_symbol_top` re-synthesised + implemented with the same B4 XDC fix applied (false-path AXI-Lite boundary + 10% I/O budget):

Metric	M1 (comb. risk, old XDC)	M2 (B4 XDC)	Change
CLB LUTs	19,942	19,827	−115
CLB Registers (FF)	8,984	9,320	+336 (risk pipeline)
RAMB36	4	4	unchanged
RAMB18	8	4	−4
DSPs	0	0	—
WNS	−5.362 ns	−0.433 ns	+4.93 ns ✓
Failing endpoints	10,899	2	−10,897 (99.98% fewer)
Total on-chip power	2.869 W	2.804 W	−0.065 W

M2 closes the multi-symbol datapath. The 2 residual failing endpoints are FF→OBUF paths (same OOC clock-insertion artifact as B4); both are on the `m_axis_*` AXI-Stream output boundary, not the book→strategy→risk trade path. In the real AWS F1 Shell build those ports are registered by the Shell and will close in-context.

Summary of what is confirmed closed at 200 MHz:

- All internal register→register paths (the full tick-to-trade pipeline: parse → BRAM book → strategy → risk → decision FF) meet 200 MHz in both `tick_to_trade_top` and `multi_symbol_top`.
- 4-symbol BRAM book scales correctly: 4 symbols → 8 BRAM tiles, < 2% LUT utilisation on VU9P.
- Residual OOC violations are I/O boundary artifacts; they do not exist in the in-context F1 build.

Run history

Run	Change	WNS	Status
B0	registered FF order-book array	−2.373 ns	FAIL (route-bound)
B1	<code>entries[]</code> → BRAM	−4.563 ns	FAIL (combinational chain)
M1	multi-symbol ×4 (BRAM, comb. risk)	−5.362 ns	FAIL (chain + sym mux)
B2	register risk/decision stage	−2.217 ns	FAIL I/O only — datapath closed ; 87 endpoints
B3	pipeline strategy (2-stage) + collar ref	TBD	RTL done (205 ns sim, 95/95); <i>re-run synth</i>
B4	XDC: false-path AXI-Lite + 10% I/O budget	−0.578 ns	FAIL FF→OBUF only (47 eps); internal logic MEET
M2	multi-symbol + B4 XDC	− 0.433 ns	FAIL 2 eps (OOC I/O artifact); datapath CLOSED
B5	XDC: false-path all CL/SH boundary ports	TBD	constraints done; <i>re-run synth</i>